

(1). $\lim_{n \rightarrow \infty} S_n = \text{Finite} \rightarrow C$ (2). $\lim_{n \rightarrow \infty} S_n = +\infty$ or $-\infty \rightarrow D$

(3). $\lim_{n \rightarrow \infty} S_n = \text{not a unique limit} \rightarrow (0)$

(4). For G.S. $1 + r + r^2 + \dots$

- (i). Converges if $|r| < 1$
- (ii). Diverges if $r \geq 1$
- (iii). Oscillates if $r \leq -1$

(5). P-series $\Rightarrow \frac{1}{1^p} + \frac{1}{2^p} + \dots$

- (i). Converges if $p > 1$
- (ii). Diverges if $p \leq 1$

(6). D'Alembert's Ratio Test.

$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = l$

(i). Convergence $\rightarrow l < 1$

(ii). Diverges $\rightarrow l > 1$

(7). Vn Test $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = l$ {a finite quantity}

$(P\text{-series or G.S.}) \vee_n \Leftrightarrow u_n \rightarrow v_n = \frac{1}{n^{p-q}}$

$p \rightarrow \text{Deno.}$
 $q \rightarrow \text{Num.}$

(8). CAUCHY'S ROOT TEST

$\lim_{n \rightarrow \infty} (u_n)^{1/n} = l$

(i). converges $\rightarrow l < 1$

(ii). diverges $\rightarrow l > 1$

(9). RAABE'S TEST $\Rightarrow$

$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = l$

(i). C $\rightarrow l > 1$

(ii). D $\rightarrow l < 1$

(10). Logarithmic Test $\Rightarrow$

$\lim_{n \rightarrow \infty} \left(n \log \left(\frac{u_n}{u_{n+1}} \right) \right) = l$

(i). converges $l > 1$

(ii). diverges $l < 1$

(11). ALTERNATING SERIES. $u_1 - u_2 + u_3 - u_4 + \dots$

is such that

(i). $u_1 > u_2 > u_3 \dots$

(ii). $\lim_{n \rightarrow \infty} u_n = 0 \Rightarrow$ series converges.

& if $\lim_{n \rightarrow \infty} u_n \neq 0 \Rightarrow$ oscillatory.

(1). $\cos z = \frac{e^{iz} + e^{-iz}}{2}$

$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$

$\tan z = \frac{e^{iz} - e^{-iz}}{i(e^{iz} + e^{-iz})}$

(3). $\sinh x = \frac{e^x - e^{-x}}{2}$

$\cosh x = \frac{e^x + e^{-x}}{2}$

$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

(3). $\sin(ix) = i \sinh x$

$\cos(ix) = \cosh x$

$\tan(ix) = i \tanh x$

(4). $\sinh(ix) = i \sin x$

$\cosh(ix) = \cos x$

$\tanh(ix) = i \tan x$

$$(5). \sinh^{-1}(w) = \log(w + \sqrt{w^2 + 1})$$

$$(6). \cosh^{-1}(w) = \log(w + \sqrt{w^2 - 1})$$

$$(7). \tanh^{-1}(w) = \frac{1}{2} \log\left(\frac{1+w}{1-w}\right)$$

$$(1). D^n(e^{ax+b}) = a^n e^{ax+b}$$

$$(2). D^n(a^x) = (\log a)^n a^x$$

$$(3)(i). D^n(ax+b)^m = m(m-1)\dots(m-n+1) a^n (ax+b)^{m-n} \quad (n < m)$$

$$(n > m)$$

$$(ii). D^n(ax+b)^m = 0$$

$$(iii). D^n\left(\frac{1}{ax+b}\right) = \frac{(-1)^{n-1} (n-1)! a^n}{(ax+b)^{n+1}}$$

$$(5). y = \cos(ax+b)$$

$$y_n = a^n \cos\left(ax+b + \frac{n\pi}{2}\right)$$

$$(4). y = \log(ax+b)$$

$$y_n = \frac{(-1)^{n-1} (n-1)! a^n}{(ax+b)^n}$$

$$(6). y = \sin(ax+b)$$

$$y_n = a^n \sin\left(ax+b + \frac{n\pi}{2}\right)$$

$$(7). y = e^{ax} \sin(bx+c)$$

$$y_n = x^n e^{ax} \sin(bx+c+n\phi)$$

$$r = (a^2 + b^2)^{1/2} \quad \phi = \tan^{-1}(b/a)$$

(8). LEIBNITZ'S THEOREM.

$$D^n(uv) = D^n(u) \cdot v + {}^nC_1 \cdot D^{n-1}(u) \cdot D(v) + {}^nC_2 \cdot D^{n-2}(u) \cdot D^2(v) + \dots + {}^nC_n \cdot D^n(u) \cdot v$$

(9). MACLAURIN SERIES.

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

(10). TAYLOR'S THEOREM.

$$f(a+h) = f(a) + h f'(a) + \frac{h^2}{2!} f''(a) + \dots$$

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots$$

- (1). Length of Tangent $\Rightarrow \frac{y}{y'} \sqrt{1+y'^2}$
 " " Normal $\Rightarrow y \sqrt{1+y'^2}$
 " " Subtangent $\Rightarrow y/y'$
 " " Subnormal $\Rightarrow y \cdot y'$

(2). Differential coeff. of Length of Arc (Cartesian Form).

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\frac{ds}{dy} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}$$

(ii). Parametric form $\Rightarrow x = f(t) \quad y = \phi(t)$.

$$x = r \cos \theta \quad y = r \sin \theta$$

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

(3). ANGLE B/w R.V & Tangent to a Polar Curve $\Rightarrow$
 $\tan \phi = r \cdot \frac{d\theta}{dr}$
 $\phi \rightarrow$ +ve if it is measured in anticlockwise.

(4). ANGLE OF INTERSECTION OF 2 CURVES.

$$\tan(\phi_1 \wedge \phi_2) = \frac{\tan \phi_1 \wedge \tan \phi_2}{1 + \tan \phi_1 \tan \phi_2} = \frac{m_1 \wedge m_2}{1 + m_1 m_2}$$

(5). POLAR SUBTANGENT $\Rightarrow r^2 \frac{d\theta}{dr}$.

" SUBNORMAL $\Rightarrow \frac{dr}{d\theta}$.

(6). LENGTH OF $\perp$ FROM POLE ON A TANGENT $\Rightarrow$
 $p = r \sin \phi$
 $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta}\right)^2$

(7). PEDAL Eqⁿ { Relation b/w p & r } radius vector.
 Length of $\perp$.

Case-1 For Cartesian eqⁿ $\Rightarrow f(x, y) = 0$.
 Eqⁿ of tangent at $(x_1, y_1) \Rightarrow (Y - y) = \frac{dy}{dx} (X - x) = 0$

$$p = \frac{x \frac{dy}{dx} - y}{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}$$

$$\text{also } r^2 = x^2 + y^2.$$

Case-2 For POLAR CURVE.

$$\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta}\right)^2$$

(1). For CART. CURVE. R.O.C $\rho = \frac{\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{3/2}}{d^2y/dx^2}$

$$\rho = \frac{\left\{1 + \left(\frac{dx}{dy}\right)^2\right\}^{3/2}}{d^2x/dy^2}$$

(2). PARAMETRIC $\rho = \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''}$

(3). Polar form. $r = f(\theta)$. $r_1 = \frac{dr}{d\theta}$. $r_2 = \frac{d^2r}{d\theta^2}$.

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{(r^2 + 2r_1r_2 - r_1r_2)}$$

(4). PEDAL CURVES. $\boxed{\rho = r \cdot \frac{dr}{dp}}$ $\boxed{\rho = r \sin \phi}$

Asymptotes.

(1). $C = -\frac{\phi_{n+1}(m)}{\phi'_n(m)}$

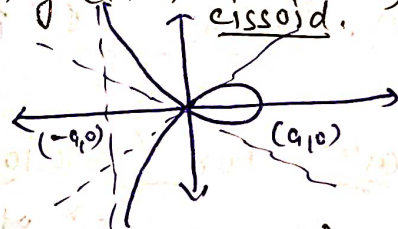
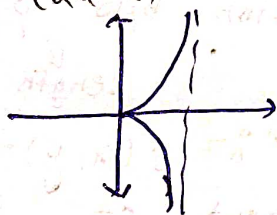
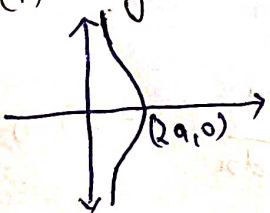
(2). $\frac{c^2}{2!} \phi_n''(m) + c \phi'_{n-1}(m) + \phi_{n-2}(m) = 0$

$\star \frac{c^3}{3!} \phi_n'''(m) + \frac{c^2}{2!} \phi_{n+1}''(m) + \frac{c}{1!} \cdot \phi_{n-2}'(m) + \phi_{n-3}(m) = 0$

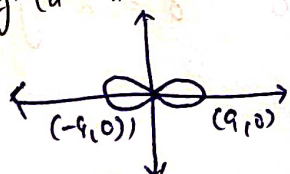
(1). $xy^2 = 4a^2(2a-x)$

(2). $y^2(2a-x) = x^3$

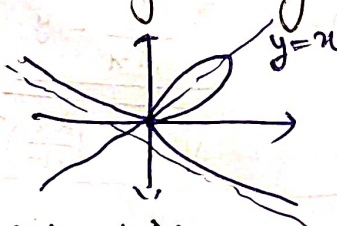
(3). $y^2(a+x) = x^2(a-x)$ cissoid



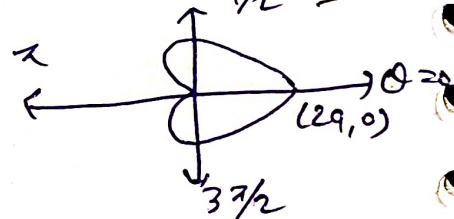
(4). $y^2(a^2+x^2) = x^2(a^2-x^2)$



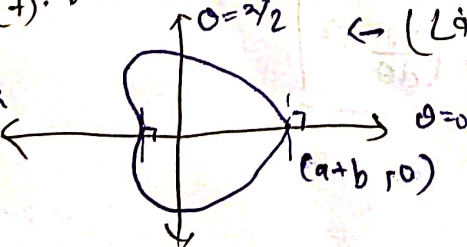
(5). $x^3 + y^3 = 3axy$



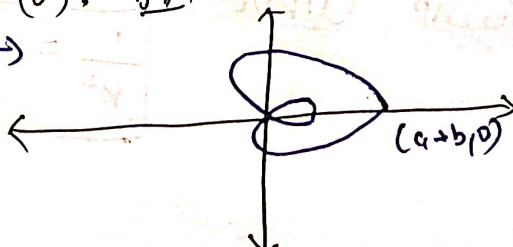
(6). $r = a(1 + \cos \theta)^{1/2}$ cardioid



(7). $r = a + b \cos \theta$ ($a > b$)



(8). $b > a$



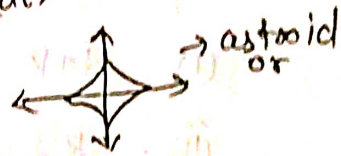
(9). $r^2 = a^2 \cos(2\theta)$ (Bernoulli).

(10). $r = a \sin 2\theta$

(1). $s = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

(2). $s = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

(3). $s = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$



(1). $s = \int_{x_1}^{x_2} 2\pi y \sqrt{1 + y'^2} dx \rightarrow$ about x-axis

(2). $s = \int_{y_1}^{y_2} 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \rightarrow$ about y-axis

(3). $s = \int_{\theta_1}^{\theta_2} 2\pi y \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$ (4).

(1). $V = \int_{x_1}^{x_2} \pi y^2 dx \rightarrow$ about x-axis

(2). $V = \int_{y_1}^{y_2} \pi x^2 dy \rightarrow$ about y-axis

(3). $V = \int_{\theta_1}^{\theta_2} \pi y^2 \frac{dx}{d\theta} d\theta$ (4).

(1). $\text{Rank}(A) = \text{Rank}(C) \Rightarrow$ system is consistent.
 (a). if $\text{Rank}(A) = \text{Rank}(C) = n \rightarrow$ (no. of unknowns) $\Rightarrow$ have unique soln.

(b). if $\text{Rank}(A) = \text{Rank}(C) < n \Rightarrow$ infinite solns.

(2). $\text{Rank}(A) \neq \text{Rank}(C)$ system is inconsistent & has no soln.

(3). $(A - dI) = 0 \rightarrow$ characteristic ~~Matrix~~ polynomial of A .

$$AX = dX$$

$$(A - dI)X = 0$$

* product of eigen values = $|A|$
 * sum " " " = $\text{tr}(A)$

(4). If d is a char. value of a sq. matrix A , then

(i). $d+k$ is a ch. value of matrix $(A + kI)$.

(ii). $k d$ " " " " " (kA) .

(iii). $\frac{1}{d}$ " " " " " $A^{-1} \Rightarrow \boxed{\frac{|A|}{d} \rightarrow \text{adj}(A)}$

(iv). d^2 " " " " " A^2 .

(5). Eigen values are same for A & A^T .